Kaduna.lk

R26-SE-026

Project Proposal Report

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Intelligent 3D Accident Claim System

B.Sc. (Hons) Degree in Information Technology Specialized in Software Engineering

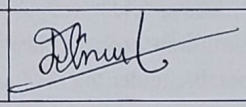
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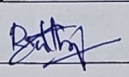
March 2026

Declaration

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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Abstract

The motor insurance sector in Sri Lanka is currently facing a critical trust deficit regarding digitized claim systems. While mobile-based photo submissions exist, insurance providers remain hesitant to adopt full automation due to significant vulnerabilities: poor visibility in low-light accident scenarios, the inability to verify the driver's state (e.g., intoxication), and the prevalence of "claim recycling," where historical damage is submitted as new. Furthermore, existing solutions lack a cohesive data protection framework, leaving sensitive policyholder data vulnerable under the Sri Lanka Personal Data Protection Act (PDPA) No. 9 of 2022. This research proposes a novel, integrated platform that utilizes advanced Image Understanding and Processing (IUP) 3D reconstruction to modernize the claim lifecycle. At its core, the system will allow the user to capture multi-angle photos and convert standard 2D mobile captures into a 3D and 360-degree view of the accident scene. This approach allows for automated lighting enhancement to resolve nighttime visibility issues and provides a volumetric assessment that is significantly harder to spoof than isolated 2D images. To combat fraud, the platform integrates temporal metadata verification ensuring the claim is filed by the authorized driver at the true time of the incident and location. Architecturally, the solution is built on a microservices framework to ensure seamless ecosystem integration between the vehicle owner, insurer, and authorized repair centres. A "Privacy-by-Design" (PbD) approach is adopted to enforce PDPA compliance, utilizing data minimization and Role-Based Access Control (RBAC) to ensure sensitive PII is only accessible to authorized stakeholders. By shifting the focus from simple digitization to high-fidelity image processing and legal-technical compliance, this research provides a robust, transparent, and secure foundation for the future of Insurance Tech in Sri Lanka.

Keywords: 3D Scene Reconstruction, Insurance Tech, Microservices, Sri Lanka PDPA, Insurance Fraud Detection, Privacy-by-Design, Image Understanding and Processing, Low-light Enhancement.

Acknowledgment

I would like to express my sincere gratitude to my project supervisor for their continuous guidance, constructive feedback, and invaluable support throughout the development of this research. Their insightful comments and professional expertise greatly contributed to improving the quality and direction of this project.

I am also deeply thankful to the academic staff of the department for providing the necessary knowledge, resources, and encouragement that helped me successfully complete this study. Their lectures and technical guidance laid the foundation for the concepts applied in this project.

My appreciation extends to the industry professionals, insurance representatives, garage owners, and spare parts dealers who shared their practical insights and experiences. Their contributions were essential in identifying real-world challenges and shaping the proposed solution.

I would also like to thank my colleagues and friends for their support, discussions, and motivation during the research and development phases. Their encouragement helped me overcome various challenges encountered throughout this journey.

Finally, I am especially grateful to my family for their unwavering support, patience, and understanding. Their continuous encouragement has been the driving force behind my academic and professional achievements.

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List of Abbreviations

RBAC	Role Based Access Control
PDPA	Personal Data Protection Act
PII	Personally Identifiable Information
NIC	National Identity Card
AES	Advanced Encryption Standard
API	Application Programming Interface
HTTPS	HyperText Transfer Protocol Secure
PbD	Privacy-by-Design
TLS 1.2	Transport Layer Security version 1.2
SfM	Structure-from-Motion
JWT	JSON Web Tokens
IUP	Image Understanding and Processing

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Introduction

1.1Background and Context

The motor insurance industry in Sri Lanka faces a critical trust and efficiency gap in accident management. While basic digitization has started, current "photo-and-upload" systems are highly vulnerable to fraud. Insurers struggle with "claim recycling" where users upload historical damage as new and the inability to verify the driver's state (e.g., intoxication) at the time of the incident. Furthermore, poor visibility in nighttime accidents often renders 2D images useless for accurate assessment. Beyond fraud, existing workflows are legally vulnerable; they lack a robust technical framework to comply with the Sri Lanka Personal Data Protection Act (PDPA) No. 9 of 2022, risking sensitive policyholder data during transmission between fragmented stakeholders.

1.1.1 Who is Affected and Why It Matters

The proposed system targets several key stakeholders in the Sri Lankan motor insurance ecosystem. Over 1,000,000 private vehicle owners often experience delayed insurance claim processing and potential risks to their personal data privacy. Insurance companies, including 14 licensed providers in Sri Lanka, face financial losses due to fraudulent or staged claims and incur high operational costs from manual accident inspections. In addition, more than 181 registered repair centres and towing service providers rely on limited 2D images to estimate vehicle damage, which can lead to inaccurate assessments. Therefore, improving the accident documentation and claim verification process is both economically important for reducing insurance costs and socially significant for protecting user data privacy.

1.1.2 Analysis of Existing Solutions and Limitations

Current accident reporting solutions mainly rely on basic messaging platforms such as WhatsApp or simple 2D web portals, which have several limitations. Standard photos provide no depth perception, as they capture flat images that cannot accurately represent the distance or depth of vehicle damage. These systems are also vulnerable to tampering, since image metadata can be modified, allowing old accident photos to be submitted as new claims. In addition, many existing applications struggle with low visibility, as they lack image-processing capabilities to improve photos taken in low-light or poor weather conditions. There is also a compliance gap, because general-purpose platforms do not follow a privacy-by-design approach and therefore fail to enforce PDPA requirements such as data minimization and purpose limitation. Furthermore, current systems are siloed, meaning they either focus only on visual evidence or only on fraud analytics, without providing a unified and legally compliant platform that connects multiple stakeholders.

1.1.3 Proposed Solution Overview

This research proposes an Intelligent 3D Reconstruction of accident site and Privacy-Enforced Microservices Ecosystem: The proposed solution introduces several key features to improve accident documentation and claim verification. First, the system uses 3D and 360° scene reconstruction, where a guided mobile application collects multiple photos from different angles and processes them using a Structure-from-Motion (SfM) photogrammetry pipeline to generate a volumetric 3D model of the accident scene. This model provides better spatial understanding of vehicle damage compared to normal images. The system also applies low-light image processing to improve photos captured during nighttime accidents. In addition, a fraud mitigation mechanism uses temporal and biometric verification to help confirm that the authorized driver is reporting a genuine incident. Finally, the system follows a PDPA-compliant microservices architecture**,** using Role-Based Access Control (RBAC) to ensure that each stakeholder, such as insurers and garages, can only access the specific data required for their role.

This research focuses on the design and implementation of the 3D Reconstruction Service for the Sri Lankan motor insurance context. While the broader platform envisions a multi-stakeholder ecosystem, this dissertation specifically contributes the technical demonstration of how PDPA compliance and privacy-enforced data delivery can be embedded directly into a reconstruction pipeline an aspect largely missing in existing mobility and insurance solutions. The service covers guided multi-angle image capture, low-light enhancement, Structure-from-Motion based 3D model generation, and role-based access control ensuring each stakeholder receives only the data their role permits.

1.1.4 Relevant SDG Alignment

SDG 9: Industry, Innovation, and Infrastructure support the digital transformation of Sri Lanka’s insurance sector by introducing advanced technologies such as high-tech 3D reconstruction and “Privacy-by-Design” microservices. SDG 11: Sustainable Cities and Communities enhance urban emergency response by enabling instant accident scene digitization, creating a safer and more transparent ecosystem for vehicle owners and pedestrians. Meanwhile, SDG 16: Peace, Justice, and Strong Institutions strengthen legal and institutional integrity by supporting the enforcement of the Sri Lankan PDPA No. 9 of 2022, ensuring that data processing within institutions remains transparent, accountable, and resistant to fraud.

1.1.5 Research Cluster Alignment

Cybersecurity and Data Privacy focus on engineering decentralized architectures that utilize API Gateways to technically enforce national data protection laws. At the same time, Distributed Systems and Cloud Computing involves implementing scalable microservices capable of managing complex interactions among multiple stakeholders within a high-load environment.

1.2 Literature Review

Recent research highlights the increasing use of computer vision and data analysis techniques in automobile insurance claim processing. Gupta et al. [1] provide a comprehensive survey of artificial intelligence and computer vision methods used for vehicle damage assessment. Their study explains how deep learning, image analysis, and automated damage detection systems are being applied to assist insurance companies in evaluating accident claims more efficiently. The survey also highlights the potential of advanced imaging techniques, including multi-view analysis, to support more accurate accident documentation.

Several studies have focused on generating three-dimensional representations of accident scenes from two-dimensional images. Somnuk and Kanchanasut [3] proposed a method that uses a convolutional neural network to reconstruct vehicle accidents from multiple 2D images. Their approach allows investigators to better understand the spatial relationship between vehicles and analyse the impact of collisions. Similarly, Kholodkov et al. [4] presented a technique for reconstructing accident sites using photogrammetry and spatial modelling methods. This approach enables investigators to create a realistic 3D representation of the accident environment, improving visualization and analysis compared to traditional photo-based documentation.

In contrast, fraud detection research mainly focuses on analysing claim data rather than visual reconstruction. Schrijver et al. [2] conducted a systematic literature review on automobile insurance fraud detection methods, highlighting the use of data mining, machine learning, and anomaly detection techniques to identify suspicious claim patterns. While these approaches aim to detect fraudulent claims, 3D reconstruction methods focus primarily on improving the accuracy and completeness of accident evidence used in claim evaluation.

Building on these industry trends, the technical implementation of such a platform requires a shift away from traditional, rigid software structures toward a Microservices architectural style. While microservices are rapidly gaining popularity for their ability to break complex systems into independent, manageable services [5], this transition is not without its challenges. Research into the “pains and gains” of this approach suggests that while the gains in scalability and deployment speed are significant, the operational complexity and the gap between academic theory and industrial practice must be carefully managed [6]. This architectural flexibility is particularly vital when handling sensitive data across multiple stakeholders. For instance, the use of an extended Role-Based Access Control (RBAC) model originally proven effective in securing distributed healthcare platforms provides a robust blueprint for managing the diverse access needs of vehicle owners, insurers, garages and marketplaces while preventing unauthorized data leaks [7]. Also, a decentralized microservices structure allows for the meaningful implementation of data minimization and purpose limitation; by isolating data into specific service boundaries, the system can technically operate with much less data than traditional centralized models, directly aligning with modern legal requirements like the Sri Lankan PDPA [8].

1.2.1 Existing research approaches

Many existing studies use computer vision techniques to automatically detect vehicle damage from accident images, with some approaches analyzing multiple images taken from different angles to improve damage detection accuracy by examining several viewpoints. Another research direction involves reconstructing accident scenes using images captured from different perspectives, while other studies focus on identifying fraudulent insurance claims through data mining and machine learning techniques. In addition, modern system design emphasizes data minimization, where data-driven systems are engineered to process only the absolute minimum amount of information required for a transaction rather than relying solely on procedural policies. This aligns with the Privacy-by-Design (PbD) engineering approach, where recent research, particularly by Biega and Finck, suggests that privacy should be treated not as a manual policy but as a technical constraint embedded directly into the system architecture.

1.2.2 Existing Products and Services

Many Sri Lankan insurance companies still rely on legacy monolithic web portals, which, while stable, are slow to update and difficult to scale. In contrast, global AI-based platforms leverage computer vision to estimate vehicle damage directly from photos. Locally, some insurers provide mobile apps for claim reporting, but these typically serve as simple front ends to a large, centralized database rather than offering fully integrated, scalable solutions.

1.2.3 Limitations of Existing Systems

In many developing countries, older monolithic insurance systems face significant challenges. Data sprawl is common, as user PII (Personally Identifiable Information) is often shared across entire databases, making it difficult to comply with the PDPA's "Purpose Limitation" rule. Compliance gaps are another issue, as most systems were built before the enactment of the Sri Lanka PDPA (2022) and lack features such as automated "Right to Erasure" or granular consent management at the service level. Centralized architectures also create security bottlenecks, where a breach of the main database exposes all user data, from NIC numbers to bank details. Additionally, most systems only allow users to upload 2D photos or videos, which capture the accident from limited angles, and many accidents occur at night or in low-light conditions, producing images with noise, blur, or poor lighting that obscure critical details like vehicle damage or road conditions. Users can upload images in any format or angle, and without guided capture processes, photos are often inconsistent in quality, further complicating accident documentation and analysis. While some research approaches improve accident analysis and fraud detection, they generally operate independently, focusing either on visual accident reconstruction or data-driven fraud detection, rather than integrating both techniques into a comprehensive solution for insurance claim verification.

1.2.4 Summary of Related Work

Domain	Source	Approach	Key Finding	Limitation
Vehicle Damage Assessment	Gupta et al. (2024) [1]	Deep learning, multi-view image analysis	AI improves damage detection accuracy	No 3D reconstruction or PDPA compliance
Insurance Fraud Detection	Schrijver et al. (2024) [2]	Data mining, anomaly detection	Detects suspicious claim patterns	No visual or spatial verification
3D Accident Reconstruction	Somnuk & Kanchanasut (2021) [3]	CNN with multi-view images	Reconstructs accidents from photos	No low-light enhancement or metadata validation
Accident Site Reconstruction	Kholodkov et al. (2024) [4]	Photogrammetry, spatial modelling	Generates realistic 3D scenes	No claim system integration or privacy controls
Microservices Architecture	Dragoni et al. (2017) [5]	Service-based system design	Improves scalability and flexibility	Theory-practice gap in implementation
Microservices Challenges	Soldani et al. (2018) [6]	Grey literature review	Identifies operational complexity	Does not consider legal compliance or privacy protection requirements
Role-Based Access Control	Ferreira & Cruz-Correia (2021) [7]	Extended RBAC model	Controls multi-user data access	Applied mainly to healthcare and not directly applied to insurance platforms
Data Minimization & Privacy	Biega & Finck (2022) [8]	Privacy-by-Design framework	Emphasizes technical privacy enforcement	Mostly theoretical and lacks practical implementation

Table 1 - Summary of Related Work

1.3 Research Gap

The PDPA Compliance Gap While the Personal Data Protection Act No. 9 of 2022 has established a new legal baseline for Sri Lanka, most existing motor insurance software solutions remain non-compliant. Current platforms often operate on legacy monolithic architectures where personal data such as NIC numbers, biometric facial data, and precise accident locations is stored in a single, accessible database. This creates a significant "Compliance Gap," as these systems lack the technical mechanisms to enforce Data Minimization or Purpose Limitation automatically. There is a documented lack of research into localized microservices frameworks that utilize an API Gateway to physically intercept and block unauthorized data access between third-party stakeholders like garages and insurers.

Systems in Sri Lanka are primarily 2D-based. These are insufficient because they cannot verify the spatial depth of an accident. Insurers face high rates of "claim recycling" (using old damage photos for new claims) and are unable to assess volumetric damage depth from flat images. Furthermore, the "Nighttime Visibility" remains a major hurdle; standard 2D captures in low-light conditions often fail to show structural damage, forcing insurers to default to manual, paper-based inspections which delay the process and increase costs.

1.3.1 Why Current Approaches are Insufficient

Existing local systems are primarily 2D-based, making them highly vulnerable to insurance fraud and "claim recycling." Because standard photos lack depth and volumetric data, it is easy for users to submit old accident photos as new incidents, a problem compounded by poor visibility in nighttime captures. Current software lacks the image-understanding capabilities to verify the temporal and spatial depth of an accident scene, forcing insurers to rely on manual, slow inspections that delay the entire claim lifecycle for legitimate policyholders.

Furthermore, current platforms lack technical integration with the Sri Lanka PDPA No. 9 of 2022. Most systems use monolithic databases where sensitive personal data is shared across all stakeholders' insurers, garages, and dealers without granular control. This "Data Sprawl" violates the principles of data minimization and purpose limitation. Without a distributed microservices architecture to enforce privacy as a technical constraint, these platforms remain legally vulnerable, as they cannot automatically "mask" or restrict sensitive information based on the specific role of the person accessing the data.

1.3.2 Novelty of the system

This research proposes a novel Intelligent 3D Reconstruction and Privacy-Enforced Microservices Ecosystem designed to address both fraud and privacy challenges in insurance claims. The system leverages 3D and 360-degree imaging, with a mobile application guiding users to capture the vehicle from multiple viewpoints to generate a volumetric 3D image. This approach makes fraudulent claims nearly impossible, as the depth of dents must correspond to the physics of the reported incident. To handle accidents in low-light conditions, the platform incorporates image enhancement to reconstruct nighttime scenes, effectively "adding light" to ensure clear damage assessment. Architecturally, the ecosystem employs a Microservices-based Privacy-by-Design framework, where the API Gateway acts as a legal firewall, technically restricting access for example, a garage service can only see the anonymized 3D damage report needed for repair estimates, not the policyholder's personal ID. Additionally, anti-fraud verification is enforced through Temporal Metadata, ensuring that 3D models are generated only from photos taken in real-time at the verified GPS coordinates of the accident, preventing practices like "claim recycling."

Objective

2.1 Main Objective

To design a privacy-aware microservices architecture for the Sri Lankan motor insurance ecosystem and implement key components that use 3D and 360 image reconstruction and image understanding to reduce fraud while supporting compliance with the Personal Data Protection Act (PDPA) No. 9 of 2022**.**

2.2 Specific Objectives

1. Guided Capture The mobile app uses the device gyroscope and accelerometer to guide the user in capturing a minimum of 15 photographs with 70% overlap. The shutter locks until the correct angle is confirmed, ensuring sufficient photo coverage for reliable reconstruction.
2. Low-Light Enhancement Zero-DCE runs directly on the device before upload, restoring nighttime accident photos without requiring any training dataset. This solves the visibilitygap that currently forces insurers into slow manual inspections.
3. 3D Reconstruction Enhanced images are processed on a cloud server - OpenMVG generates a sparse point cloud, OpenMVS densifies and meshes it, and the original photos are projected to produce a fully textured OBJ file.
4. Fraud Validation GPS coordinates and timestamps are captured server-side at upload, independently of EXIF data which can be spoofed on Android. Damage depth in the mesh is cross-checked against an impact physics table - any mismatch automatically raises a fraud flag.

Methodology

3.1 Research Approach

This research follows the Design Science Research (DSR) methodology, which focuses on the creation and evaluation of innovative IT system to organizational problems. Given the unique challenges of the Sri Lankan insurance market specifically high fraud rates and new PDPA legalities the DSR approach allows for an iterative process of designing a 3D reconstruction and a privacy-preserving microservices architecture, followed by rigorous testing against real-world insurance constraints.

3.2 Technical Methods and Algorithms

Stage 1

Guided Multi-Angle Capture: The React Native mobile application uses device sensor data accelerometer and gyroscope to guide the vehicle owner in capturing a minimum of 20 photographs, approximately 5 positions per side of the vehicle. Each consecutive photo must share at least 60% frame overlap with the adjacent capture, with 70% as the working target. The shutter is locked until the gyroscope confirms the correct angular displacement from the previous shot, typically 10–15 degrees of rotation at a camera-to-vehicle distance of 1.5–2 meters. If fewer than 12 valid frames are confirmed, the app prevents submission and highlights missing positions using an overlaid vehicle silhouette guide. If 12–19 photos are received, the system attempts a partial reconstruction and flags the output as Low-confidence, automatically routing the claim for manual inspection rather than rejecting it outright, ensuring no legitimate claim is blocked due to a technical capture failure.

Stage 2

Low-Light Enhancement: Before reconstruction, the system detects low-light conditions using image brightness analysis. Zero-DCE is selected as the enhancement model over EnlightenGAN for three reasons: it is unsupervised and requires no paired training dataset, its model size is under 80KB making it suitable for on-device preprocessing on the mobile app before upload, and it was designed specifically for real-world low-light conditions without reference images.

Stage 3

3D Point Cloud Generation: The reconstruction pipeline runs four sequential stages on a cloud server. First, OpenMVG generates a sparse point cloud by detecting and matching feature points across all photos. Second, OpenMVS densifies this into a detailed point cloud via depth map computation. Third, a Poisson surface reconstruction algorithm converts the dense cloud into a polygonal mesh. Fourth, the original photographs are projected onto the mesh to produce a photorealistic textured surface, exported as an OBJ file. Processing is fully asynchronous. The mobile app uploads photos and immediately receives a job ID, Azure Service Bus queues the reconstruction job, and a push notification is sent when the model is ready. The target end-to-end time is under 3 minutes. The final output is rendered in a WebGL viewer within the insurance officer portal. A textured mesh is required rather than a raw point cloud - insurance officers need a photorealistic surface to assess damage severity, and garages need visible surface detail to estimate repair costs.

Stage 4

Temporal & Fraud Validation: Since EXIF metadata embedded in JPEG files is editable on Android devices and cannot be trusted alone, the system captures metadata server-side at upload time. The mobile app transmits the device timestamp and GPS coordinates as separate HTTP request headers now of upload, independently of image EXIF data. If the EXIF-embedded timestamp and the server-recorded upload-time value differ by more than 10 minutes, the claim is automatically flagged. Gallery uploads are disabled entirely - only camera captures initiated within the app session are accepted - preventing submission of pre-existing photographs. For damage consistency analysis, each damage region identified in the mesh is assessed by extracting maximum depth displacement in millimeters and the surface area of deformation, forming a damage signature vector. This vector is cross-referenced against an impact physics lookup table: minor impact under 15 km/h produces deformation under 8 mm depth; moderate collision produces 15–40 mm; structural damage exceeds 40 mm over a large surface area. If the reported incident type does not match the corresponding signature range, a fraud flag is raised. This heuristic is grounded in vehicle damage assessment literature, specifically Gupta et al. [1].

Stage 5

RBAC-Masked Delivery: The finalized 3D model and associated claim data are passed through the API Gateway, which applies the authenticated user's RBAC role to mask or expose specific data fields. Vehicle owners receive claim status and a 360° view only. Insurance officers receive the full textured mesh and fraud score. Garages receive an anonymized damage report with no access to personal identifiers or claim metadata, technically enforcing PDPA data minimization at the service level rather than relying solely on procedural controls.

Tools and Platforms

Category	Tool	Purpose	Justification
Mobile / Image Capture	Smartphone Camera (User Phone)	Users take accident photos from different angles.	Smartphones are widely available and allow users to quickly capture photos at the accident site without extra equipment.
Mobile Application	React (React Native)	Provides the mobile interface for users to upload photos and report accidents.	React allows cross-platform mobile development and can guide users to capture photos from correct angles.
Backend Service	Node.js + Python and Keras	Runs backend microservices that handle requests, accident reports, and data processing.	Node.js supports scalable microservices and handles multiple requests efficiently.
System Orchestration	Kubernetes (K8s)	Manages and scales backend services automatically.	Kubernetes helps run multiple services reliably and allows the system to scale when demand increases.

3D Processing	OpenMVG / AliceVision	Converts multiple accident photos into a 3D or 360-degree accident model.	These tools support photogrammetry and can generate depth information from images to analyze vehicle damage.
Messaging / Event Processing	Azure Service Bus	Handles communication between services and manages system events.	Provides durable, auditable asynchronous messaging between microservices
Security & Access Control	Role-Based Access Control (RBAC) + JWT, RBAC, AES-256, TLS 1.3	Controls which users can access system features and data and encrypt data	RBAC ensures only authorized users (insurers, garages, vehicle owners) can access sensitive information.

Table 2 - Tools and platforms

3.3 Validation Strategy and Performance Metrics

The system's effectiveness will be evaluated across several key dimensions. 3D Reconstruction Accuracy will be assessed by comparing damage measurements generated by the platform against those recorded by licensed insurance inspectors on the same vehicles, using metrics such as Mean Absolute Error (MAE) and correlation coefficients. PDPA Compliance will be tested through automated security assessments that attempt to access PII via unauthorized RBAC roles, with the gateway's ability to block such access measured against a target of 100%. Finally, Performance Under Load will be evaluated by subjecting the system to 2,000 concurrent requests, representing the estimated peak daily load, to verify response times and the effectiveness of auto-scaling mechanisms.

3.4 Data Collection and Ethical Considerations

Data collected through stakeholder interviews (vehicle owners, garages, insurers) and system usability testing. All collected data will be anonymized, securely stored, and used strictly for research purposes. User consent will be obtained prior to participation, ensuring compliance with data privacy and ethical standards.

High-Level System Architecture

Microservices-Based Platform Architecture and Component Design

The proposed solution is a microservices-based unified platform that connects key automotive stakeholders including vehicle owners, insurance companies, garages, spare parts sellers, and towing services through a 3D accident reconstruction module at its core.

The system adopts a microservices architecture with a centralized API Gateway, where each major business function is implemented as an independent service. This design ensures scalability, flexibility, and maintainability across the entire platform.

All client applications, including the Unified Mobile Application and the Insurance Officer Web Portal, communicate with the backend exclusively through the API Gateway. The gateway is responsible for request routing, authentication validation, security enforcement, and access control, acting as the single entry point for all system interactions.

User authentication is managed by a dedicated Authentication Service, which issues JSON Web Tokens (JWT) upon successful login. The API Gateway validates these tokens before granting access to any downstream service, ensuring that only authorized users can interact with the system.

Microservices communicate using two approaches depending on the nature of the operation. Synchronous communication via internal REST APIs is used for operations requiring immediate responses, such as dispatching towing services. Asynchronous event-driven communication via Azure Service Bus is used for operations that can be processed independently, such as notifications, service record updates, and predictive maintenance tasks.

Each microservice follows the Database-per-Service pattern, maintaining its own dedicated encrypted database. This ensures service isolation, prevents cross-domain data leakage, and allows each service to scale its storage independently.

Together, these architectural decisions produce a scalable, secure, and loosely coupled platform capable of supporting accident response, vehicle service coordination, and automotive marketplace operations within a single unified ecosystem.

Microservices Architecture Compare with other Architectures

Criteria	Layered	SOA	Microservices	EDA
Multi-organization integration	Low	Medium	High	Medium
Independent scalability	Low	Medium	High	High
API-first design	Medium	High	High	Medium
Security isolation	Low	Medium	High	Medium
Implementation complexity	Low	High	Medium	High
Suitability for marketplace	Low	Medium	High	Medium

Table 3 - Microservices Architecture Compare with other Architectures

How the 3D Accident Model is Generated

The 3D accident model generation process begins when the vehicle owner captures multiple photographs of the damaged vehicle using the guided mobile application. The app uses device sensor data to direct the user to capture images from the specific angles required for an accurate reconstruction, ensuring consistent and sufficient photo coverage before submission.

Once uploaded, the system assesses the lighting conditions of the captured images. If the photos were taken in low-light or nighttime conditions, the enhancement pipeline automatically improves brightness and clarity using Zero-DCE, ensuring that damage details remain visible and usable for accurate assessment.

The enhanced images are then processed by the Structure-from-Motion engine, which identifies common feature points across all photographs and calculates the spatial relationship between them. This allows the system to understand the precise shape of the vehicle surface and reconstruct the geometry of the damaged area in three dimensions.

Using this spatial data, the system generates a fully textured 3D model of the damaged vehicle, capturing the depth, structure, and surface detail of the damage far more accurately than any single two-dimensional photograph could provide.

Finally, the generated model is cryptographically bound to the device timestamp and GPS coordinates recorded at the time of capture. This temporal anchoring verifies the exact time and location of the incident, making it technically impossible to submit pre-existing or recycled photographs as a new insurance claim.

Microservices Architecture System diagram of components

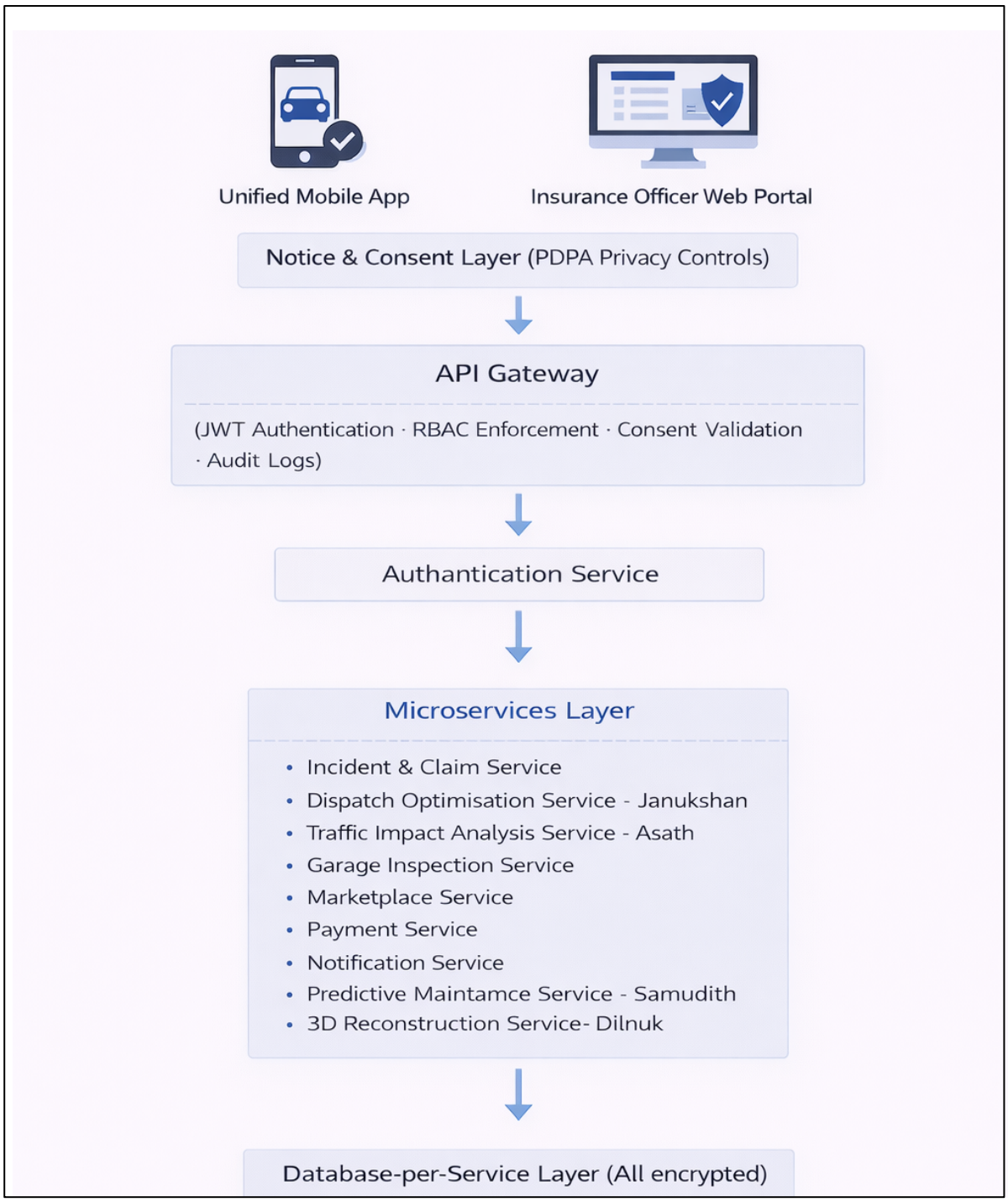


Figure 1 - System diagram of components

3D Reconstruction Service System Diagram

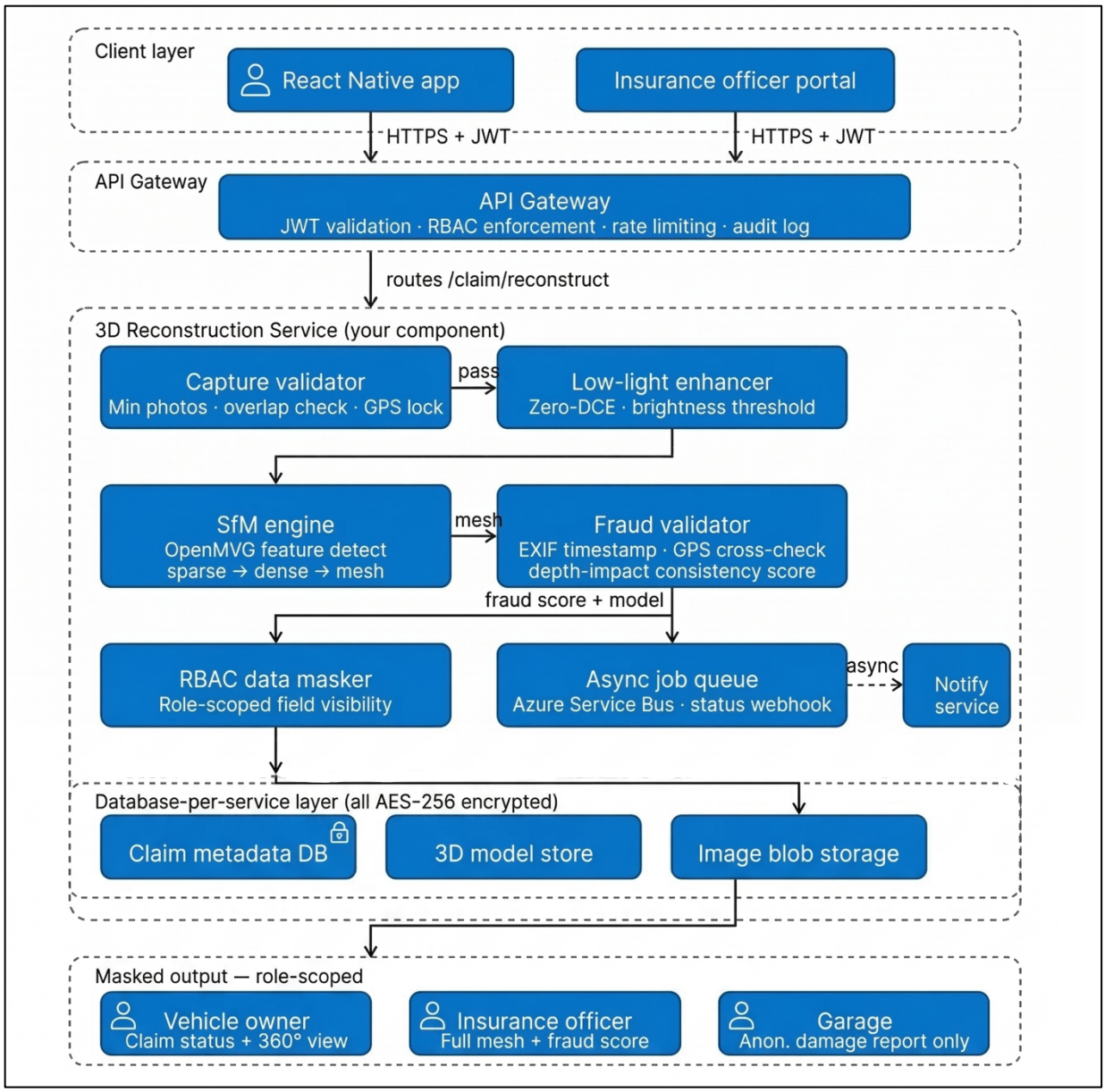


Figure 2 - 3D Reconstruction service system diagram

4.1Scalability Considerations

Based on estimated usage statistics in Sri Lanka, the system is expected to process approximately 100–1000 service requests per day during early adoption. Considering an estimated 600,000 young vehicle owners, around 100 accidents and 100 vehicle breakdowns may occur daily, resulting in roughly 200 incidents per day. If each incident triggers about 10 system actions, the system may handle approximately 2,000 requests per day. Even with a 10× growth in usage, the system would need to support around 20,000 requests per day.

To support this growth, the proposed system uses a containerized microservices architecture deployed with Docker, enabling horizontal scalability while maintaining system modularity and simplicity.

4.1.1 Independent Service Scalability

Each functional domain is implemented as an independent microservice. This design allows services to scale independently based on workload demand. For example, the Marketplace Service can scale during periods of high spare-part searches, while dispatch or towing services can scale during periods of increased accident frequency.

4.1.2 Fault Isolation

Separating services ensures that failures remain localized. For example, issues within the Towing or Location Service will not affect the Garage Inspection or Marketplace Services, thereby improving overall system availability and reliability.

4.1.3 Database-Per-Service Pattern

Each microservice maintains a dedicated database to enforce loose coupling, data encapsulation, and independent scaling of storage systems. This approach prevents cross-domain data leakage and supports independent database scaling.

4.1.4 API Gateway Traffic Management

The API Gateway is responsible for distributing requests across multiple service instances, filtering, routing, and rate-limiting incoming traffic, and caching frequently requested responses to reduce database load.

4.2 Security and PDPA Compliance Considerations

The system follows a Privacy-by-Design approach to comply with Sri Lanka's Personal Data Protection Act (PDPA).

4.2.1 Authentication and Authorization

JWT-based authentication is used for all users. Tokens contain encrypted role information and enable stateless authentication suitable for distributed systems.

4.2.2 Role-Based Access Control (RBAC)

The system's privacy and security architecture centers on the API Gateway as the primary enforcement mechanism. Consent verification middleware ensures user permissions are respected at every interaction, and data minimization within services ensures that each microservice processes only the information strictly necessary for its function.

4.2.3 Data Protection Mechanisms

All communications use HTTPS with TLS 1.2+. Databases use AES-256 encryption, and media files such as inspection photos are stored in encrypted storage. Passwords are hashed using bcrypt or Argon2, and no plaintext credentials are stored at any point in the system.

4.2.4 Data Retention and Deletion Policies

To comply with the PDPA, personal data is retained only for defined operational periods with automated retention policies in place. Users may request deletion of their personal records where legally permissible. Service record data is retained according to applicable regulatory or contractual requirements.

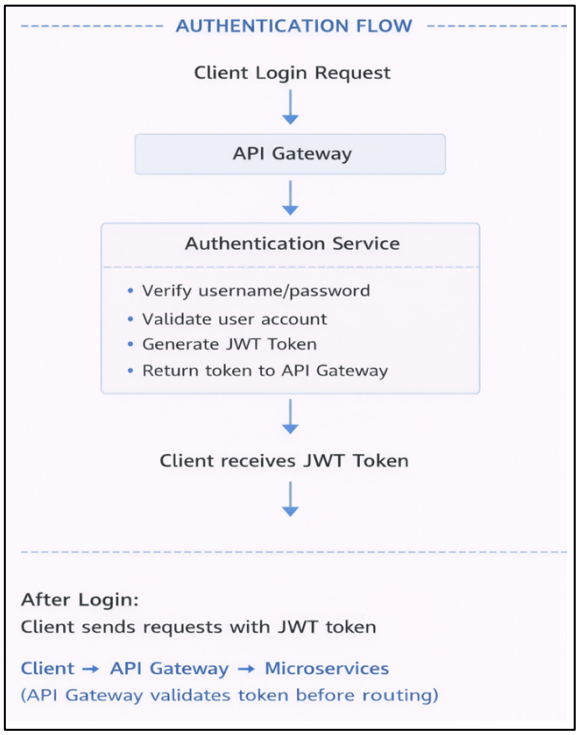


Figure 3 - Authentication flow

User Requirements

5.1 Stakeholder identification

5.1.1 Primary Stakeholders

The primary stakeholders of the system are private vehicle owners, representing an estimated 700,000 main target users, along with insurance companies, registered garages and repair centers, spare parts dealers, and towing service providers.

5.1.2 Secondary Stakeholders

The secondary stakeholders include system administrators, auto advertisers, and regulatory authorities where required.

5.2 Functional requirements

ID	Feature	Description
FR-01	Guided Multi-Angle Capture	The mobile app shall use sensor data to guide users in capturing overlapping photos from angles required for SfM reconstruction.
FR-02	Low-Light Detection & Enhancement	The system shall detect dark conditions and apply light enhancement-based restoration before reconstruction.
FR-03	3D Scene Reconstruction	The system shall synthesize uploaded image sets into a volumetric 3D and 360° model.
FR-04	Temporal Metadata Anchoring	Each 3D model shall be cryptographically bound to its EXIF timestamp and GPS coordinates to prevent reuse.
FR-05	Automated Fraud Scoring	The system shall calculate a fraud risk score from depth analysis and metadata discrepancy checks.
FR-06	PDPA-Compliant API Gateway	The gateway shall mask PII fields based on the authenticated user's RBAC role in real time.
FR-07	Consent & Rights Management	Users shall be able to view, manage, and withdraw data sharing consents per PDPA requirements.

Table 4 - Functional requirements

5.3 Non-functional requirements

ID	Requirement	Description
NFR1	Role-Based Access Control (RBAC)	Only authorized insurance officers shall be able to view financial data, while garages shall only access the 3D repair model and necessary vehicle specifications.
NFR2	Data Encryption	All Personally Identifiable Information (PII) and 3D models must be encrypted using AES-256 when stored and TLS 1.3 when transmitted to prevent data breaches.
NFR3	Concurrent Handling	The microservices architecture must support at least 100 concurrent claim submissions without performance degradation, utilizing cloud auto-scaling capabilities.
NFR4	Elderly Accessibility	The mobile user interface must include high-contrast visual elements and simplified interaction to ensure usability for senior citizens or stressed drivers during an accident.
NFR5	Offline Resilience	The application must allow local photo capture and metadata storage in areas with poor internet connectivity, automatically syncing with the 3D reconstruction service once the connection is restored.

Table 5 - Nonfunctional requirements

Commercialization Plan

6.1 Target market and customer persona

The primary target market is private vehicle owners in Sri Lanka who regularly interact with insurance companies, garages, and spare parts dealers. The core customer persona is aged between 25 and 55, owns a private vehicle, uses insurance services, and values convenience, transparency, and time-saving digital solutions. Secondary customers include spare parts dealers and automotive businesses seeking digital exposure.

6.1.1 Value Proposition

The platform provides one integrated solution for accident reporting, repair management, spare parts purchasing, and breakdown assistance. It offers transparent pricing, reduced fraud risk, faster insurance claim processing, and convenient towing and nearby garage discovery all accessible from a single application.

6.1.2 Revenue Model

The platform follows three revenue streams. The commission-based model provides free marketplace access initially to attract users and dealers, with a small commission introduced later on spare parts sold through the platform. Advertisement revenue is generated by allowing automotive businesses to display paid advertisements within the application. Future premium subscriptions will offer optional paid plans for garages and dealers to promote their services and spare parts to vehicle owners.

6.1.3 Cost Estimation

The key cost categories include system development covering software development and testing, cloud hosting and infrastructure, ongoing maintenance and security updates, and marketing and user acquisition expenses.

6.1.4 Pricing Strategy

Vehicle owners receive free access to the platform. Spare parts dealers are onboarded at no cost initially, with a gradual introduction of a 3–5% commission on transactions as the platform matures. Competitive advertising rates will be offered to automotive businesses seeking visibility within the application.

6.1.5 Competitive Advantage

The platform is the first integrated multi-stakeholder automotive ecosystem in Sri Lanka, combining insurance, garage, towing, and spare parts marketplace functionality within a single system. It offers real-time coordination, price transparency, GPS-based emergency support integration, and 3D reconstructed accident scene images for insurance assessment — capabilities not available in any existing local solution.

6.1.6 Intellectual Property Considerations

The platform's intellectual property encompasses the custom-developed system architecture and integration model, the proprietary database structure and marketplace workflow, potential registration of software copyright, and trademark registration for platform branding in the future.

Budget And Justification

The following table outlines the estimated costs for developing the system. All costs are estimated in Sri Lankan Rupees (LKR).

Category	Item Description	Estimated Cost (LKR)
Software	OS, IDEs, Docker Desktop, and Database Tools and Modeling API Credits	50,000
Cloud Services	Azure Hosting & Managed Database Services	45,000
APIs	Google Maps API	25,000
Human Effort	Development, Research, and Testing (Approx. 800 hours)	600,000
Total		720,000 LKR

Table 6 - Estimated cost table

7.1 Software Licenses

To minimize costs, the system utilizes open-source technologies including Visual Studio Code, Docker Community Edition, Node.js, Python, and PostgreSQL. Using industry-standard open-source tools ensures the system remains cost-effective, highly maintainable, and compatible with modern DevOps practices.

7.2 Cloud Services

Azure cloud hosting is necessary to provide multi-stakeholder access, as Insurance Officers, Garages, and Vehicle Owners operate from different physical locations and require a live deployed server. Cloud services also enable the use of Managed Databases, which provide the encryption and automated backup features required for PDPA compliance.

7.3 APIs and AI Services

The Google Maps API is required for the location tracking feature, enabling the system to accurately pinpoint accident sites as a core requirement of the Incident Service.

7.4 Human Effort Cost

The estimated human effort reflects a single student developer working approximately 30-40 hours per week over 8 months. This accounts for the complexity involved in designing and implementing the 3D reconstruction pipeline, integrating the privacy-enforced microservices architecture, developing the guided mobile capture module, and conducting validation testing against real-world insurance inspection measurements.

Work Breakdown Structure

WP	Task	Responsible	Timeline	Deliverables
WP1	Data collection and preparation	Dilnuk De Silva	Month 1-2	Find Research Papers
WP2	Create the Architecture	Dilnuk De Silva	Month 3-4	System architecture diagram and design document
WP3	3D accident reconstruction and 360 image generation	Dilnuk De Silva	Month 4-7	Working prototype to generate 3D/360 accident models
WP4	Fraud detection module development and app screens development	Dilnuk De Silva and Members	Month 7-9	Fraud detection model and testing results
WP5	System integration and evaluation	Dilnuk De Silva and Members	Month 9-10	Final integrated system, performance evaluation, and project report

Table 7 - Work breakdown structure

Gantt Chart

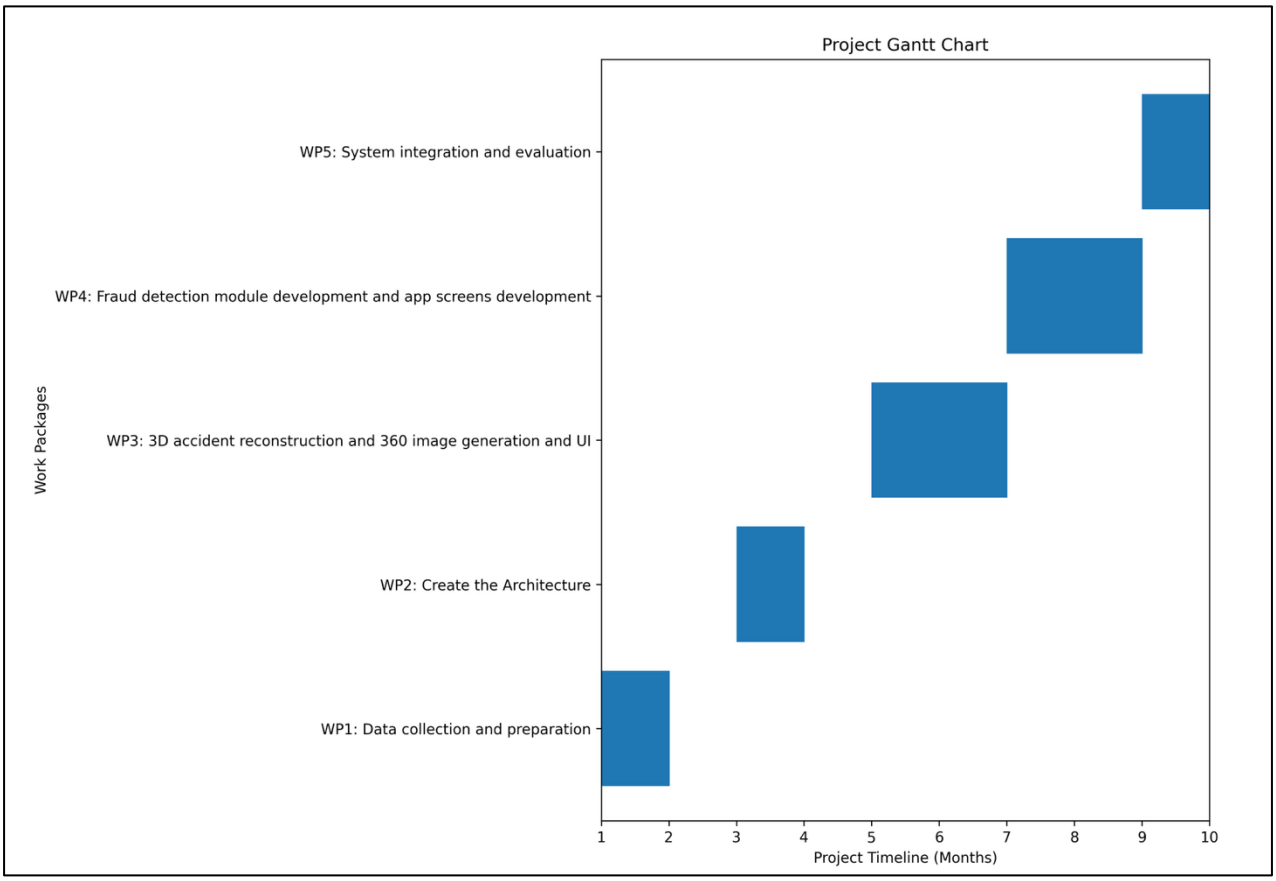


Figure 4 - Gantt chart

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Data Collection

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List of Auto body shops in Sri Lanka

[11] Number of motor car insurance companies
Insurance Regulatory Commission of Sri Lanka

[12] Find what people really need in Sri Lanka
Google Sheet Vehicle Owners

[13] Find what approaches taken by Insurance Companies in Sri Lanka
Google Sheet Insurance Companies

[14] Sri Lanka's Personal Data Protection Act, No. 9 of 2022 (PDPA)
https://www.parliament.lk/uploads/acts/gbills/english/6242.pdf

Appendices

Appendix A - Survey Instruments (Problem Validation)

Target Audience: Vehicle Owners & Insurance Officers

Sample Questions for Vehicle Owners:

1. On a scale of 1-10, how satisfied are you with the current speed of accident claim processing in Sri Lanka?
2. What is the claim process that you faced after the accident?
3. What are the problems you face during that process from accident happen to get the car repaired and get the claim to your account?
4. Would you prefer an automated AI-based damage estimate over waiting for a manual inspector? (Yes/No)
5. Did you find spare parts for your vehicle? and did the insurance pay the claim for brand new spare parts?

Sample Questions for Insurance Officers:

1. What is the average time taken from Accident Reported to visit the accident?
2. What is the process when getting claim for the accident?
3. Do you have online system and WhatsApp to use in an accident?
4. What are the main challenges you face in this process?
5. How often do disputes occur regarding the initial damage cost estimation?

Appendix B - Preliminary UI Mockups

1. Vehicle Owner Mobile App (Find spare parts):
Features: The Spare parts details will be shown in the application with a search button, The user will search for the spare parts in the marketplace. The system will show the available dealers.

2. Vehicle Owner Upload Picture:
Features: The user will have to hold the phone in the angle that the system request and then images will be taken, the shutter will be disabled until the phone is in correct angle.

3. Vehicle Owner Upload Picture:
Features: After taking picture the app will show a next button that will straight upload the photo. The app will keep a timestamp and the location of the photo and will not allow to use the phone memory for photos.

Appendix C - Risk Analysis

Risk Type	Description	Impact	Likelihood	Mitigation Strategy
Security	Data breach at the API Gateway level.	High	Low	Implement Rate Limiting and JWT validation at the Gateway; use TLS between services.
Compliance	Failure to delete user data across all services (PDPA violation).	High	Medium	Use an Event-Driven architecture to ensure “Eventual Consistency” for data deletion across all DBs.
Operational	Resistance from garages to use a new digital platform.	Medium	High	Ensure the Garage UI is extremely simple and provides value (faster payments).

Appendix D - Ethical Documents & PDPA Compliance

The project will strictly adhere the Sri Lanka PDPA No. 9 of 2022.

Informed Consent Form: Before using the app, every user will be presented with a clear “Privacy Notice” explaining:
This notice clearly explains what data is collected, including NIC numbers, location coordinates, and accident photographs, who the data will be shared with such as the assigned insurance officer and the specific garage handling the repair, and how the user may withdraw their consent at any time through the application settings.